Multi-Level Set Estimation: Formalizing the Confidence Box and Straddle Acquisition Function

1 Introduction

In the context of multi-level set estimation, the objective is to classify a continuous input space based on multiple unknown objective functions. Specifically, given m objective functions $f_1, f_2, \ldots, f_m$, the goal is to identify the region where all objectives simultaneously exceed a specified threshold. Assuming the thresholds are all zeros, the target multi-level set is defined as $\{x \in \mathcal{X} \mid f_i(x) \geq 0,\ \forall i = 1, \ldots, m\}$.

This document formalizes the concept of a *confidence box* for an input point x and derives a boundary-seeking acquisition function, extending the Straddle heuristic to $\mathbb{R}^m$.

2 Marginal Confidence Intervals

Assuming the unknown objective functions are modeled using Gaussian Processes (GPs) or a Multi-output GP, the posterior predictive distribution for each objective $f_i(x)$ at a point x provides a predictive mean $\mu_i(x)$ and standard deviation $\sigma_i(x)$.

The marginal confidence interval $C_i(x)$ for the i-th objective is defined as:

$$C_i(x) = [l_i(x), u_i(x)] \tag{1}$$

where the lower and upper bounds are given by:

$$l_i(x) = \mu_i(x) - \beta_i^{1/2}\sigma_i(x) \tag{2}$$

$$u_i(x) = \mu_i(x) + \beta_i^{1/2}\sigma_i(x) \tag{3}$$

Here, β_i is a scaling parameter determining the confidence level, often derived via information-theoretic bounds.

3 The Confidence Box Formulation

The confidence box $B(x)$ projects the uncertainty at point x into the $\mathbb{R}^m$ objective space. It is defined as the Cartesian product of the m individual confidence intervals:

$$B(x) = C_1(x) \times C_2(x) \times \cdots \times C_m(x) = \prod_{i=1}^{m} [l_i(x), u_i(x)] \tag{4}$$

Expressed as a set of vectors $\mathbf{y} \in \mathbb{R}^m$, the confidence box is:

$$B(x) = \{\mathbf{y} \in \mathbb{R}^m \mid \mathbf{l}(x) \leq \mathbf{y} \leq \mathbf{u}(x)\} \tag{5}$$

where $\mathbf{l}(x) = [l_1(x), \ldots, l_m(x)]^\top$ and $\mathbf{u}(x) = [u_1(x), \ldots, u_m(x)]^\top$, with inequalities applied element-wise.

4 Application to the Zero Threshold

The classification of x depends entirely on the position of $B(x)$ relative to the origin $\mathbf{0}$ and the hyperplanes $y_i = 0$ in the objective space:

- **Confident Inclusion:** x is confidently inside the multi-level set if the entire box lies in the positive orthant: $\forall i,\ l_i(x) \geq 0$.
- **Confident Exclusion:** x is confidently outside if the box strictly violates the threshold for at least one objective: $\exists i$ s.t. $u_i(x) < 0$.
- **Ambiguity (Unclassified):** x remains unclassified if $B(x)$ intersects the threshold boundaries without strictly meeting the exclusion criteria.

5 Formalizing the Straddle Acquisition Function

To efficiently sample points that reduce boundary ambiguity, we extend the Straddle heuristic into $\mathbb{R}^m$. The acquisition function takes the form $a(x) = U(x) - \lambda D(x)$, prioritizing points with high uncertainty (U) balanced against a penalty (D) indicating how the threshold splits the confidence box.

5.1 The Uncertainty Term $U(x)$

$U(x)$ represents the total volume of the confidence box $B(x)$:

$$U(x) = \mathrm{Vol}(B(x)) = \prod_{i=1}^{m} (u_i(x) - l_i(x)) \tag{6}$$

5.2 The Volume Partitions

The box $B(x)$ is partitioned into $B_1(x)$ (points where all $y_i \geq 0$) and $B_2(x)$ (the remainder). The volume of $B_1(x)$, which lies in the positive orthant, is computed by truncating the bounds at zero:

$$\mathrm{Vol}(B_1(x)) = \prod_{i=1}^{m} (\max(0, u_i(x)) - \max(0, l_i(x))) \tag{7}$$

Since $B_1(x)$ and $B_2(x)$ are mutually exclusive and comprise the entire box:

$$\mathrm{Vol}(B_2(x)) = \mathrm{Vol}(B(x)) - \mathrm{Vol}(B_1(x)) \tag{8}$$

5.3 The Straddle Penalty $D(x)$

The penalty term $D(x)$ measures boundary proximity via the absolute difference in volume between the two partitions:

$$D(x) = |\mathrm{Vol}(B_1(x)) - \mathrm{Vol}(B_2(x))| \tag{9}$$

Substituting the identity for $\mathrm{Vol}(B_2(x))$ simplifies this computation:

$$D(x) = |2\mathrm{Vol}(B_1(x)) - \mathrm{Vol}(B(x))| \tag{10}$$

5.4 Final Acquisition Formulation

Bringing these components together yields the multi-dimensional Straddle acquisition function:

$$a(x) = \mathrm{Vol}(B(x)) - \lambda\, |2\mathrm{Vol}(B_1(x)) - \mathrm{Vol}(B(x))| \tag{11}$$

6 Analytical Properties

- **Maximum Exploration:** When the threshold perfectly bisects the box ($\text{Vol}(B_1(x)) = \frac{1}{2}\text{Vol}(B(x))$), the penalty $D(x) = 0$, and $a(x)$ peaks at the total uncertainty volume $U(x)$.
- **Confident Penalty:** If the box is fully classified ($\text{Vol}(B_1(x)) = \text{Vol}(B(x))$ or 0), the penalty maximizes to $D(x) = \text{Vol}(B(x))$. Setting $\lambda \approx 1$ ensures safe regions are heavily discounted.